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Framework for Agile Transformation: Guiding Organizations Through Cultural, Structural, and Competency Shifts in Project Management

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Abstract: This paper aims to provide a framework for guiding organizations toward Agile practices for project management. Through a case study analysis conducted in an information technology company, the paper aims to explore the interplay between current organization structure and project management approaches, the set of competencies, and the cultural change required for a transition toward Agile transformation. The paper provides a framework comprising four main blocks necessary for guiding organizations toward Agile practices for successful project management. The framework proposed demonstrates that for organizations to be nimble and to improve teamwork performance, it is necessary to undertake a set of changes in culture, competencies, structure, and approaches. The findings highlight the importance of a structured approach to Agile transformation, with particular emphasis on top management support and continuous learning. By formalizing Agile initiatives through structured approaches and continuous engagement, organizations can foster a shared sense of purpose among team members, ensuring sustained commitment and facilitating smoother transitions.

Keywords: agile project management; agile transformation; project management; agility framework

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Citation: Ndou, Valentina, Alessandro Ingrosso, and Anna Di Girolamo. 2024. Framework for Agile Transformation: Guiding Organizations Through Cultural, Structural, and Competency Shifts in Project Management. *Administrative Sciences* 14: 301. https://doi.org/10.3390/admsci14110301

Received: 20 August 2024
Revised: 15 October 2024
Accepted: 7 November 2024
Published: 14 November 2024

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1. Introduction

In today's dynamic environment marked by volatility, uncertainty, complexity, and ambiguity (VUCA) (Bennett and Lemoine 2014), an increasing number of organizations are implementing Agility in their way of working (Alt et al. 2020) to achieve greater efficiency and to implement dynamism (Magistretti and Trabucchi 2024; Trabucchi et al. 2019). Originally developed as a software engineering methodology, Agile project management (APM) has gained traction across various industries (Digital AI: 15th Annual State of agile Report 2022) and is now one of the top 10 topics for organizations and managers (Kappelman et al. 2017). Unlike traditional methodologies, Agile approaches focus on iteration and learning to navigate the unpredictability of the modern world (Magistretti et al. 2023) and emphasize the capacity to swiftly adjust strategies, structures, processes, roles, skills, and technologies to capitalize on new opportunities and deliver increased value to stakeholders. Agile methodologies prioritize creativity and adaptability over rigid processes, acknowledging the unpredictability inherent in today's business landscape (Hoeseb and Tanner 2020).

Academic research has widely argued that Agile methodologies offer several opportunities for organizations. Agile teams, relying on informal communication and tacit knowledge, can develop and market systems more quickly (Mehta et al. 2023). Their flexibility and adaptability enable effective responses to changing customer demands and market conditions (Hoeseb and Tanner 2020). Continuous, iterative quality controls improve system quality and customer satisfaction, while a collaborative environment boosts employee happiness (Venkatesh et al. 2020).

Yet, despite the widespread adoption of Agile methodologies, many organizations face challenges in effectively implementing Agility due to behavioral, cultural, and team-related obstacles.

Challenges associated with Agile approach implementations are several and involve coordination issues due to the need to synchronize across multiple teams and departments (Hannay and Benestad 2010; Kalenda et al. 2018; Rolland et al. 2016). The transition is especially challenging in organizations accustomed to traditional management practices, often resulting in change resistance exacerbated by concerns over transparency and power dynamics (Conboy et al. 2018; Moe et al. 2019). It has been argued by Hobbs and Petit (2017) that integrating Agile into established organizational structures often faces resistance due to entrenched processes and technological architectures that conflict with Agile's iterative and planless approach. Also, insufficient knowledge and training in Agile concepts among management and teams hinder successful implementation (Kalenda et al. 2018), with a lack of investment in qualified coaches and proper training exacerbating these challenges.

Magistretti and Trabucchi (2024), in a recent article, argue that while Agile is a well-recognized mechanism to respond to rapid external changes, its successful implementation requires considering Agile as a Culture. This perspective emphasizes the need for a strategic orientation and human-centric attitude necessary for rapid alignment and change. Therefore, Business Agility, as a comprehensive organizational approach, necessitates a fundamental change in culture and mindset. Following this, in this paper, we aim to demonstrate the following empirically: How can Agile transformations be successfully implemented in organizations to enhance Business Agility and address behavioral, cultural, and team-related challenges?

Drawing on case study research, this paper aims to propose, test, and evaluate an evolutionary innovative framework for guiding organizations toward successful Agile practices. The framework proposed demonstrates how a company has managed successfully and gradually to shift from an inadequate and unsatisfactory traditional waterfall model for development to an Agile approach, while simultaneously scaling it up aggressively.

The paper is structured as follows: we begin with an exploration of Agile project management methodologies, along with the frameworks utilized for their implementation, and an examination of associated drawbacks and challenges. Subsequently, we present the methodology employed in this research, elucidating its intricacies and rationale. Furthermore, we meticulously present the findings and outcomes derived from this study, offering valuable insights and implications. Finally, we encapsulate our research journey with comprehensive conclusions and shed light on avenues for future development and exploration.

2. Literature Review

2.1. Agile Project Management

Since the introduction of the Agile Manifesto in 2001, researchers and practitioners have increasingly focused on Agile project management (Abrar et al. 2019) due to its benefits related to higher-quality organizational outcomes, faster time to market, higher team productivity, increased flexibility and adaptability, and improved collaboration (Kalenda et al. 2018; Hüllmann et al. 2024).

Generally speaking, Agility has been defined as a dynamic capacity to recognize and respond to constantly changing and often ambiguous requirements in complex environments. It involves overcoming challenges by redirecting organizational practices and resources into new areas and problem-solving approaches (Felipe et al. 2016).

In the field of project management, Agile is defined as (Conboy and Fitzgerald 2004) "the continual readiness of an entity to rapidly or inherently, proactively or reactively, embrace change, through high quality, simplistic, economical components and relationships with its environment".

Agile approaches and methodologies have gained popularity across diverse industries, including banking, telecommunications, education, retail, and construction (Limaj and Bernroider 2022).

These methods aim to ensure flexibility in the face of changes and uncertainties, accelerate the development process, and emphasize communication and human connection as crucial for managing projects in a VUCA (volatile, uncertain, complex, ambiguous) context (Alt et al. 2020). The literature presents various interpretations of Agility: as a method (Hobbs and Petit 2017), as a capability to thrive in a competitive and unpredictable environment by responding quickly and effectively to changes (Nejatian et al. 2018), and as a perspective for learning and reacting to change while enhancing customer value (Mikalsen et al. 2018). Adopting and implementing Agile methodologies involves an organizational change process that can occur at different levels and in various forms (Nejatian et al. 2018; Fuchs and Hess 2018). It is important to distinguish between "doing agile" and having an "agile mindset". "Doing agile" involves introducing Agile approaches solely to align processes with Agile values and principles, often leaving other organizational functions and development processes unchanged. This can create disconnected ways of working within the company, leading to inefficiencies and potential failure of the Agile initiative. Conversely, having an "agile mindset" (doing + being Agile) involves transforming the entire organization, including its culture, bureaucracy, and fundamental purpose, fostering a holistic and sustainable approach to Agility (Olsson et al. 2012).

So, if Agile adoption changes the way work is performed, Agile transformation changes the way of being. Agile transformation is a highly challenging process. Challenges of the Agile adoption process may be related to the development process, customer issues, developer and management issues, organizational issues, technical issues, human-related issues, or inadequate and dysfunctional training (Limaj and Bernroider 2022; Fuchs and Hess 2018). Different practices and strategies may be used to address the identified specific issues (Limaj and Bernroider 2022; Mikalsen et al. 2018) and to overcome the identified problems (Gandomani et al. 2013; Nerur et al. 2005; Gandomani and Nafchi 2015).

The literature on Agile transformations presents a nuanced exploration of various facets within the realm of Agile methodologies. Case studies, such as Scheerer et al. (2014), offer insights into coordination approaches within Agile teams, shedding light on the dynamics of team coordination in Agile environments. Additionally, researchers like (Paasivaara and Lassenius 2014) have scrutinized specific measures aimed at supporting Agile transformations, contributing practical strategies for the successful implementation and sustainability of Agile practices.

Furthermore, holistic characterizations of large-scale Agile transformations have been subject to investigation. Fuchs and Hess (2018) provide comprehensive insights into the complexities involved in scaling Agile methodologies across entire organizations, offering valuable frameworks and perspectives for orchestrating such transformations effectively.

In parallel, practitioners and consultants have significantly contributed by offering various Agile frameworks and models. Examples include the Scaled Agile Framework (SAFe), Large-Scale Scrum (LeSS) (Vodde and Larman 2014), and Disciplined Agile Delivery (DAD) (Ambler and Lines 2012). These frameworks provide structured approaches to implementing Agile methodologies at scale, addressing diverse organizational needs.

Singh et al. (2013) define organizational Agility as the ability of firms to sense environmental changes and respond appropriately, emphasizing the importance of balanced sensing and response capabilities within Agile organizations. Additionally, Worley et al. (2014) advocate for Agility as a strategic advantage, enabling organizations to respond effectively to changing circumstances and maintain sustained high performance. Their framework for assessing organizational Agility offers a structured approach grounded in organization design theory and empirical studies.

Overall, the literature underscores the significance of effective coordination, tailored measures for success, comprehensive frameworks for large-scale transformations, and the

strategic advantage of organizational Agility in navigating the complexities of modern business environments.

2.2. Challenges to Agile Transformation

Despite the widespread perception that transitioning to Agile is a simple and smooth step, the reality presents a series of obstacles and complexities. At the heart of these obstacles lies the dichotomy between traditional methodologies, which often have a more structured and rigid approach, and the Agile approach, which emphasizes flexibility, adaptability, and rapid response to changes. These divergences are not only technical but also cultural, as they imply a profound shift in values, expectations, and work modalities.

The transition to Agile methodologies presents a complex array of challenges, encompassing various facets of organizational culture, individual mindset, managerial dynamics, and structural processes (Gandomani and Nafchi 2015; Singh et al. 2013). These challenges can be categorized as follows (Figure 1):



Figure 1. Agile transformation challenges (own elaboration).

1. People-Centric Challenges: Agile transformation fundamentally revolves around people, making it susceptible to a range of human-centric challenges:
    - Lack of Knowledge: Many individuals within organizations may lack a comprehensive understanding of Agile methodologies, hindering their ability to effectively implement Agile practices.
    - Cultural Barriers: Organizational cultures rooted in traditional hierarchical structures and resistance to change pose significant obstacles to Agile adoption.
    - Resistance to Change: Human nature often resists change, leading to skepticism, reluctance, and even opposition to Agile transformation initiatives.
    - Wrong Mindsets: Deeply ingrained mindsets and beliefs about project management may clash with Agile principles, impeding the adoption of Agile methodologies.
    - Lack of Effective Cooperation: Agile methodologies emphasize collaboration and teamwork, yet achieving effective cooperation among diverse stakeholders can be challenging.
2. Perception Challenges: Perception challenges stem from individuals' attitudes, beliefs, and expectations regarding Agile transformation:
    - Concerns and Worries: Individuals may harbor concerns about the implications of Agile transformation on their roles, responsibilities, and job security.
    - Limited Understanding: Despite excitement about Agile, many individuals may possess only a surface-level understanding of its principles and practices.
    - Skepticism: Past experiences of failed transformation attempts may breed skepticism and cynicism, undermining confidence in Agile methodologies.
    - Indifference or Unrealistic Expectations: Some individuals may remain indifferent to Agile transformation, while others may harbor unrealistic expectations of its outcomes.

3. Individual Behavior: Agile transformation necessitates a profound shift in individual attitudes, behaviors, and ways of working:
    - Resistance to Change: Overcoming personal resistance to change, both at the individual and team levels, is a critical aspect of Agile adoption.
    - Cultural Shifts: Embracing Agile principles requires a cultural shift within organizations, where individuals are encouraged to challenge existing norms and embrace innovation.
    - Balancing Creativity and Compliance: Fostering creativity and autonomy while adhering to Agile practices can be challenging, as individuals navigate the tension between rule-breaking and organizational alignment.
4. Managerial and Organizational Issues:
    - Agile transformation necessitates rethinking traditional managerial and organizational structures:
    - Power Dynamics: Reduced managerial authority in Agile environments may lead to resistance from existing managers accustomed to hierarchical control.
    - Structural Constraints: Aligning organizational structures with Agile principles requires overcoming structural constraints and hierarchies entrenched in traditional project management approaches.
    - Adapting to Self-Organization: Encouraging self-organization within teams while maintaining accountability and alignment with organizational goals poses managerial challenges.
5. Structural and Process Issues: Structural and process-related challenges pertain to the operational complexities of Agile transformation:
    - Scaling Agile: Scaling Agile methodologies to accommodate larger projects, distributed teams, and multi-site organizations requires overcoming logistical and coordination challenges.
    - Process Alignment: Ensuring alignment between existing processes and Agile practices necessitates careful evaluation and adaptation of organizational workflows.
    - Executive Support: Securing executive buy-in and support for Agile transformation initiatives is essential for overcoming resistance and driving organizational change.

In navigating these challenges, organizations must adopt a holistic approach that addresses cultural, behavioral, managerial, and structural dimensions of Agile transformation. Success hinges on fostering a culture of openness, collaboration, and continuous learning, where individuals are empowered to embrace change and adapt to evolving methodologies and practices.

2.3. Agile Transition

The Agile Transition Process (ATP) is a real revolution for companies, and therefore, it needs great cooperation and engagement from the main stakeholders (top and middle management, users, customers, and developers). This also means that all those involved must be able to acquire a set of transversal key skills: the ability to work collaboratively, to communicate assertively, to constantly exchange feedback, to give up competitive approaches and, finally, to adopt attitudes based on win–win logics.

Many studies pointed out that there is no single standard ATP model or best-practice example to copy and paste (Sidky 2007; Pikkarainen et al. 2014): each company represents a unique case of people, relationships, stories, contexts, regulations, and organizational culture, which together constitute a highly complex fabric. Accordingly, this fabric cannot be managed using standard approaches, and it is certainly not possible to predict which transition process would be more effective and how long would it take (Gandomani et al. 2013).

In their empirical study regarding Agile transformation, Gandomani et al. (2013) and Gandomani and Nafchi (2015) have identified two distinct fields: structural features and key activities.

Structural features are considered essential to the transformation; they make it cheaper, less challenging, and more effective.

Agile transformation should have the following structural features and lead the organization to reflect on the following points:

- It should be value-based, where "value" means value for the business. It should answer the question "Why are we adopting Agile?". It is necessary to define what the expected value will be and how it will be measured before starting the transformation.
- It should be iterative. Agile operates primarily on people's behavior, but these changes take time, effort, and commitment. It may be useful here to quote a phrase used in Toyota: "Building people before building cars".
- It should be continuous, meaning that the iterations need to be repeated continuously and without interruption, keeping in mind that Agility has no "final state" to be reached and consists of a continuous improvement: Agile transformation is a journey, not a destination.
- It should be gradual because some behaviors take more time to take root. Many companies are not immediately ready to change, and there is no need to begin with a Big Bang approach: gradual progression reduces risks and challenges and helps all those who are not ready for Agile transformation or are against it.

Agile transformation includes the following key activities, which recur at each iteration:

- Selection of practices: following the value delivery principle, the Agile practices to adopt can be sorted by priority and selected at each iteration.
- Adjustment: Agile adoption has to be sustainable, and we must remember that adjustment is not a predictive activity, because it has to do with changing people's mindset. Forcing people to change is the worst way to adopt Agile, if not the best way to fail in the attempt. Significantly increasing the transformation's chances of success is possible: it has been proven that institutionalizing the Agile adoption process and having the support of coaches and mentors helps the initiative to succeed (Sharma et al. 2022; Gandomani and Nafchi 2015).
- Assessment: this activity may help identify the challenges the team is facing and monitor the transition processes and progress. Assessment may cover various aspects such as production rate, iteration length, team size, level of cooperation, training, leadership, team self-management, business benefits, etc. Agile requires the use of new metrics aimed at measuring the benefits gained in terms of business since traditional metrics no longer work.
- Retrospective: the assessment results need to be reviewed and discussed by the team during specific retrospective meetings. Retrospective takes place on a regular basis and is aimed at identifying which improvement areas need intervention from the next iteration.
- Adaptation: the aim is to adjust and adapt to current Agile practices before adopting new ones.

Whether we are dealing with a product to develop from scratch or with a change on the run, only people can decide if they are willing to adopt an Agile approach.

The journey towards Agile methodologies represents a profound shift in both technical practices and organizational culture within companies. It goes beyond mere adoption of practices; it entails a fundamental transformation of mindset and behavior. Transitioning to Agile involves migrating from traditional development methods to more flexible and adaptive methodologies, emphasizing collaboration, adaptation, and continuous feedback.

In academia, various models have been proposed to facilitate Agile adoption within organizations. Boehm and Turner (2003) introduced a model based on five decision criteria, aiming to balance discipline and flexibility. Sidky (2007) presented a multi-phase framework

inspired by the CMMI model, emphasizing Agility measurement but facing practical applicability challenges. Jovanović et al. (2017) proposed a three-phase model focused on preparation, team-level transition, and organizational-level transition, offering a practical guide for Agile adoption.

However, these frameworks often lack empirical evaluation and fail to address crucial aspects such as training, transition management, and cultural challenges. The Scaled Agile Framework (SAFe) (Scheerer et al. 2014) addresses challenges beyond a single team but may overlook internal team dynamics and Agile adaptability exploration.

Despite their theoretical robustness, many frameworks face challenges when applied in operational contexts due to resistance to change and specific team dynamics. There is no one-size-fits-all approach to Agile transformation; companies must adopt a holistic approach, evaluating different frameworks and customizing the transition path based on their specific needs.

In this context, Agile coaches play a crucial role in guiding organizations through the transformation journey. By integrating the most effective elements from existing methodologies, a unified model of "Agile Transformation" can provide a more adaptable guide for organizations embarking on the journey towards Agility.

3. Case Study Methodology

This paper adopts a single case study methodology (Yin 2018) focusing on an IT company engaged in developing an E-suite solution. The case study revolves around the direct observation and experimentation of a tailored Agile transformation strategy within the company, facilitated by two Agile Coaches.

The case study methodology involves three main steps (Table 1):

1. Plan: Analysis of Problems Encountered;
2. Implement Proposal and Implementation of Agile Approach;
3. Evaluate Interviews with Key Team Members.

Table 1. Case study methodology.

Case Study Phases	
Plan	Analysis of Problems Encountered Time-consuming Outdated product Low collaboration among team members Unclear objective Difficult to stay in touch with changes
Implementation	Proposal and Implementation of Agile Approach Define the roles Define training needs Coaching Pilot testing
Monitor and evaluation	Interviews with Key Team Members General Manager Product Owner Scrum Master Developer 1 Developer 2 Developer 2 Agile Coach Senior 1 Agile Coach Senior 2

3.1. Case Study Description

The case described here is the company, Progetti e Soluzioni, founded in 1999, initially aimed to develop software solutions for managing services on individual demand,

specifically targeting school meals for Municipalities' Public Education Offices and contract catering companies. The flagship product, School E-Suite, has evolved significantly over the years, catering to the complex needs of over 360 Municipalities across Italy. However, its origins as a monolithic product developed using a waterfall approach have resulted in high management and maintenance costs, compounded by obsolete features.

Recognizing the limitations of traditional project management methodologies, the company embarked on a journey to explore Agile methodologies for software development.

The company decided to completely rewrite the software to create a flexible, technologically advanced, and easy-to-use solution that would take into account the customer's new needs. To this end, the company formed a working group of 10 people from different departments (sales, development, delivery, and support), which started working on the idea. The team gathered in a hotel for two weeks to focus on the project full-time and without interruption.

Then, after two months, the team gathered again for two weeks in full immersive mode to establish all the features required for the new software. The following requirements emerged from this first macro-analytical activity:

- It would have taken at least 6 to 9 months to define the technical specifications;
- Another 18–24 months for development (carried out by 5–6 developers);
- Another 6 months for testing.

It was a matter of 3 to 4 years of work until the new product was on the market, in addition to the regular maintenance activities of the School E-Suite. This estimation led the company to give up the project since it was unsustainable.

Therefore, a new working group was formed, involving this time not only key people in the company but also customers. After spending 4 days over the first month in brainstorming activities, the team made a list of over 200 items to develop. It was immediately clear that it would be hard work, but the company had gained a new sensitivity towards Agile in the meantime and therefore decided to consider this approach and these methodologies.

The first knot to loosen involved the choice of people to allocate to the new software's development. The company purchased a direct competitor who used to develop a very similar product for the same target market. The purchase effectively led to the creation of two distinct work teams competing.

We used the Scrum framework, as the most widely used Agile method (Cervone 2011), and based on its principles, new roles were assigned to the team. Scrum focuses on organizing teams according to five core values—commitment, focus, openness, respect, and courage—and identifies three main roles: Product Owner, Scrum Master, and Developer. The Scrum Teams work in iterative, time-boxed SPRINTS (short, time-boxed periods when a Scrum Team works to complete a set amount of work), beginning each with a Sprint Planning meeting to define goals and plans based on prioritized backlog items (Cervone 2011).

Hence, the company had to introduce new Developers—strangers to both teams—to form the Developer team. The Product Owner role was assigned to the group leader, who deals with the Delivery and Deployment of School E-Suite at customer sites.

She has been working for the company for over 15 years and thoroughly knows the customers and market needs and how to relate to the various stakeholders. The Scrum Master role was assigned to the new technical manager, who was recently hired by the company but already had a lot of experience as a technical director, who was responsible for the selection both of Developers and Agile Coaches. The company decided to hire the Agile Coaches because of the uncertainties surrounding the use of new tools and the acquisition of new team members. These factors might have led to proceeding by trial and error before gaining the skills needed to implement the new framework and mindset, consequently resulting in a waste of time and money.

Omnibus comes from this background: the new software solution became the pilot project for Progetti e Soluzioni SpA's Agile transition and adoption.

3.2. Agile Transformation Roadmap

To overcome all the problems encountered by the company, a tailored Agile journey framework, called ARTCO, was proposed and implemented by the company that includes 5 phases: Preliminary Analysis; Roadmap Definition; Training; Coaching; and Outcomes (Figure 2).

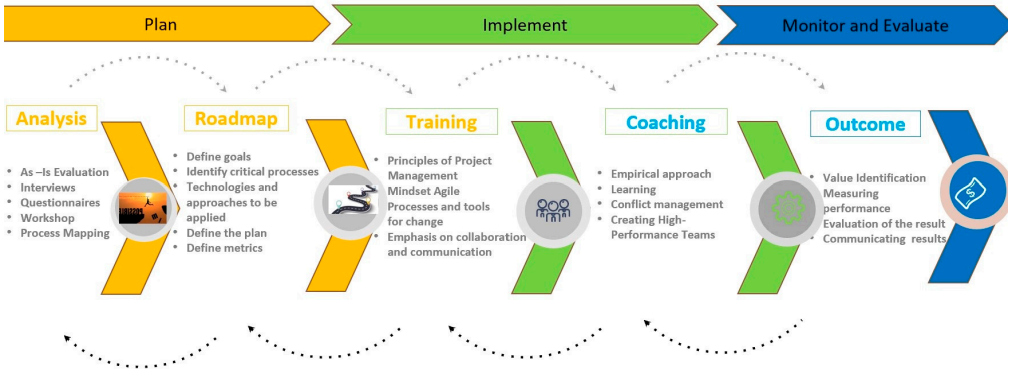


Figure 2. Iterative, adaptive, and empirical Agile transformation framework—ARTCO.

Phase 1: The analysis phase aims to examine the current situation (As-is) of the organization. This analysis is conducted through an in-depth study employing various tools such as direct observations, questionnaires, interviews, and meetings with designated managers and teams. This investigation enables a deep understanding of several elements related to planning and temporal management, the intrinsic value of projects, and dedication to customer needs.

Phase 2: The next phase involves developing a roadmap or intervention plan, a crucial step to ensure that the actions taken are targeted and effective.

Phase 3: Training phase. Training represents a fundamental pillar to ensure that all individuals involved in the transformation process are equally and effectively prepared. Training integrates a variety of elements related to interpersonal skills, such as effective communication, conflict management, self-organization, emotional intelligence, and servant leadership.

Phase 4: Coaching phase. Once training is completed, the pilot project effectively begins, initiating the coaching phase aimed at further strengthening the skills acquired during training and encouraging the practical adoption of Agile thinking. This step is crucial to ensure that the theory learned is effectively applied in a real working context, allowing for effective and lasting change. The goal of this phase is to help individuals embrace the Agile Mindset and provide them with the tools to embrace change.

Phase 5: Outcome phase. The final phase allows for the verification of the pathway in terms of business value and is extremely important: indeed, training and learning are useless if not aimed at addressing the needs of customers and product users. This phase aims to assess the team's result in terms of delivering value.

The framework is based on interconnected and evolutionary phases. Each phase of the transformation not only builds on the previous one but also interconnects with the others, creating a cohesive framework for successful Agile implementation.

Management's involvement is pivotal in driving this transformation. Their commitment fosters a culture of support and instills confidence among team members. A cultural shift is essential for effective transformation. Therefore, during the analysis phase, it is crucial to understand the current mindset and tailor the change to prioritize the organization's needs and values.

The training phase is vital for developing new competencies within teams. By focusing on interpersonal skills and Agile practices, all members are equipped to engage in self-organization and empowered decision-making. This approach reinforces a customer-centric mindset and encourages iterative thinking. Emphasizing skills like emotional intelligence

and effective communication highlights the importance of a supportive culture. As teams learn to reflect and adapt in real time, a culture of continuous improvement becomes embedded, enabling them to respond effectively to customer feedback and internal dynamics.

Additionally, the transition from a traditional hierarchy to cross-functional teams marks a significant structural evolution. This restructuring enhances collaboration and empowers teams to take ownership of their work. The effectiveness of the new structure is assessed based on the results achieved. Agile's iterative nature allows for ongoing adjustments, ensuring that the organizational structure remains aligned with project goals and customer needs.

Ultimately, the framework serves as a strategic guide, outlining the implementation of Agile principles while steering organizations through the necessary cultural, structural, and competency shifts essential for effective project management.

4. Results

4.1. Experimentation Results

The company's top management—who sponsors the initiative—was highly involved in the pilot project from the very beginning. Many studies report that top management's involvement in Agile transformation is one of the factors that significantly increase the pilot projects' chance of success (Wise and Daniel 2016; Mayner 2017). The institutionalization of the Agile transformation pilot project helps generate and maintain a high level of commitment and participation from everyone involved.

The COVID-19 pandemic forced all the stakeholders to collaborate for the first time. Given the lack of prior experiences, the impossibility of meeting in person raised various concerns, but the Agile transformation framework that was suggested nevertheless made the stakeholders feel more confident. All the people involved were also trusted and properly sensitized on the risks and opportunities of working remotely.

Secondly, the framework has facilitated frequent interactions between the stakeholders, reducing the risk of misunderstanding, enabling immediate response to changes, and encouraging a high level of cooperation.

The external constraint of remote work helped to further increase the value of the results achieved by the pilot project. The impediments of remote work—in terms of tools and methods of communication—typically increase the chance of failure of Agile transformation. But in this case, the high motivation, focus, and dedication of the people involved had to overcome the barrier of physical distance, bringing out new cooperative dynamics.

The team chose remote collaboration tools that were user-friendly either due to their ease of use or because they were already familiar to the team. The primary collaboration tools employed are described in Figure 3:

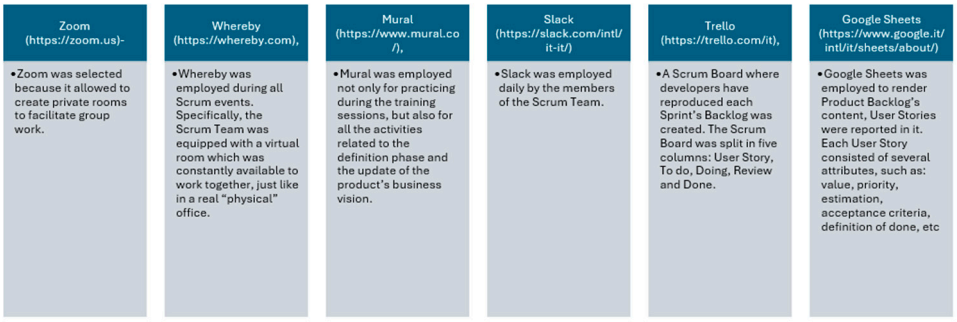


Figure 3. Collaboration tools used for the project.

The pilot project follows the following steps for Agile transition and adoption (Figure 4).

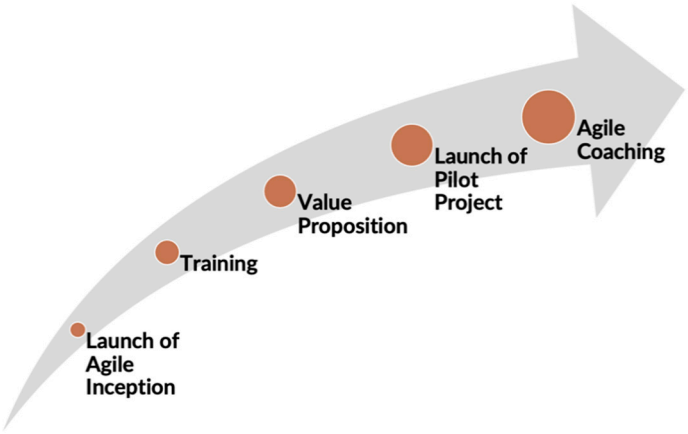


Figure 4. Project's roadmap (own elaboration).

The Omnibus pilot project started with the **Launch of Agile Inception**. The Agile transition and adaptation project named Agile Inception started with a two-hour event during which the top managers emphasized the strategic importance of mindset shift, the company's upcoming challenges and their commitment to providing support for Agile transformation. During the event, the roadmap, the team members, and the Agile Coaches were introduced. Institutionalizing the launch of the transformation motivated all the stakeholders involved, who showed themselves to be curious and eager to start working.

Training was the second step in the roadmap, and it was crucial to properly and evenly prepare all the people involved in the transformation process. Keeping training isolated and informal helps to create an environment where people—since they are free from having lots of work to do—can think, discuss, share, and start to know each other in as neutral an environment as possible.

The participants were selected to ensure role and skill diversity. Besides the project team (Scrum Team), stakeholders external to the project but within the company were involved as well. Participation was also extended to Scrum Team non-members to discuss and share Agile philosophy in those company departments that were not directly involved in the transformation but may decide to begin an Agile transformation project in the future. The primary goal was to create a shared sensitivity and to lay the foundation for a shared language. Operations staff, Developers from other departments, business figures, and the CEO took part in the training process. Six participants were directly involved in the pilot project: the Product Owner, the Scrum Master and the Developers (Scrum Team). The Scrum Team has worked together and still does at present with an active participation and a high engagement level during each Sprint. The training course consisted of two modules: Agile Mindset and Framework Scrum. The first value of the Agile Manifesto (individuals and interactions over processes and tools) determined the layout of the contents presented, which aimed to sensitize the participants on shared responsibility.

The course opened with an introduction on what complexity means and on being aware of the uncertainty of the present moment in history, when the only constant is change. This therefore set the right framework to start talking about Agile Mindset, beginning with the values and principles of the Agile Manifesto (Alt et al. 2020). The switch from the "planning and control" (predictive) to the "inspect and adapt" paradigm (adaptive or empirical) was gradually addressed. This module was supplemented with issues related to customer centricity and value-driven development. The second module aimed to make the participants familiar with the Scrum framework. Specifically, it was intended to further explore how the framework contributes to implement the empirical approach through transparency, inspection and adaptation. The module was articulated around several topics: Scrum values, roles, events, and artifacts.

The whole training course combined several elements related to the soft skills area–such as effective communication, conflict management, self-organization, emotional intelli-

gence, and servant leadership, therefore proving that an Agile Mindset consists not only of doing but also mainly of being Agile. The two Agile Coaches led the training phase together. The delivery of the course was fully remote and had an overall duration of 20 h, broken down into 5 sessions of 4 h each.

Value Proposition: Even though there was already a general product idea, the notion of Minimum Viable Product (MVP) (Ries 2011) and ongoing business requirement analyses were completely new. Defining product requirements in their entirety was the approach that had always been followed so far. Therefore, before moving to the product development phase, the Coaches scheduled a 16 h workshop in 4 sessions about creating an MVP for the value proposition. All the Scrum Team members and the CEO joined the workshop. The Agile Coaches assisted the team in exploring the value and cooperatively defining the MVP. The team proceeded also with the aid of Vision development tools such as:

- Business Model Canvas
- Impact Mapping
- Buyer Personas
- User Story

Thanks to these tools, in combination with brainstorming and cooperation techniques, the team was able to define its business scope, limited to one specific type of customer, with whom to test the product increment right from the beginning. A concrete MVP was defined, and this enabled the team to write the first User Stories which supplied the Product Backlog's first version. This approach allowed the Product Owner to identify the first potential test customers to actively involve from the very beginning, without having to wait for the entire product to be developed. This approach made it possible to immediately integrate customer and user feedback—reactively addressing their needs—and to create in a few Sprints a first product release to launch on the market to start making income. Once it tried the approach to MVP and value proposition development and User Story creation, the team was sensitized on persisting in employing these tools, being aware that—according to Agile logic—the product is constantly evolving, has no end-state, and is iteratively innovated through constant interaction with the company's internal stakeholders and support from external customers and users.

Launch of the pilot project and Agile Coaching: The previous phase enabled us to obtain a clear idea of the MVP to be carried out and to have a Product Backlog ready to be developed.

The launch of the pilot project coincided with the launch of Scrum as well. The Scrum Team agreed on a two-week duration for each Sprint. Four Sprints were arranged to perform Agile Coaching activities. At the end of this period, an overall retrospective was scheduled to evaluate the team's self-sufficiency in keeping the Agile transformation going, promoting its emancipation from the Agile Coaches as soon as possible. The pilot project then began with Sprint #1, specifically with the Sprint Planning event (Schwaber and Sutherland 2020). During this event, the Scrum Team defined the Sprint Goal, selected the User Stories (which were previously estimated and prioritized), and removed the existing ambiguities by finding a solution that could be achieved within the Sprint length. The Developers generated the Sprint Backlog and started working following the Scrum Guide's minimal set of prescriptions. The Developers worked full-time on product development, while the Product Owner and the Scrum Master were engaged for approximately 50% of their time.

The coaching activity was not full-time: for each Sprint, 16 h—broken down into four meetings—were devoted to Agile Coaching. This activity was performed during Scrum's major events:

- Sprint Planning (4 h)
- Daily (15 min)
- Sprint Review (2 h) and Sprint Retrospective (2 h)

During Planning, Review, and Retrospective events (for a total of 8 h), the Agile Coach had a double role: on one hand, he had to help the team to implement the Agile Mindset and to master Agile tools and techniques; on the other hand, he had to support it in developing cooperative attitudes and testing activities, and to help it gradually become independent and self-managed. Of particular note was the Agile Coach's contribution to the proper Daily Scrum execution. It is known that this event is often misunderstood and performed not as a planning phase but as an activity status report. That is why attending two Dailies for each Sprint was essential not only to guide the team towards achieving compliance with the 15 min time-box but also to help it understand what the purpose of the Daily was in terms of planning everyday activities and constantly verifying the Sprint Goal's achievement. After the Daily Scrum, the Agile Coach constantly stayed with the Scrum Team to help with any problem of the moment. The two four-hour biweekly meetings scheduled before the Dailies aimed at providing support and facilitating the Scrum Team in several contexts. At times, the team was supported in the Product Backlog's refinement or development. At others, facing technical questions from the Developers allowed us to find alternative solutions—so as not to jeopardize the Sprint Goal—or approaches for quality improvement. At other times, the team was coached on specific roles, such as the Product Owner and the Scrum Master. These coaching sessions outside the Scrum events have served to free people from pre-existing structures and remove the Scrum anti-patterns.

4.2. Evaluation: Interview Results

As already described, the ARTCO framework derives from the need to design a method capable of launching an Agile transformation process to carry out the Omnibus pilot project. In full accord with the empirical approach, the ARTCO framework proved effective in retrospect. To evaluate the results of the experimentation, we collected data from team members using semi-structured interviews. Each interview lasted 30 min and was conducted by the authors. All interviews were recorded and transcribed. In total, eight interviews were conducted, each with a representative of the different Scrum roles involved in the project (see Table 2).

Table 2. Interview results.

Interviews	Waterfall Method Drawbacks for the Project Development	Agile Method Advantages
General Manager Product Owner Scrum Master Developer 1 Developer 2 Developer 2 Agile Coach Senior 1 Agile Coach Senior 2	Long Development Cycles Inflexibility Limited Stakeholder Involvement Risk of Project Abandonment Lack of Adaptability High Costs and Resource Intensiveness	Customer Centricity Iterative Development Empowered Teams Flexibility and Adaptability Continuous Improvement Reduced Risk Improved Collaboration and Communication Enhanced Product Quality

Each person took part in the pilot project with different background experiences and points of view and experienced the change differently. Despite being linked by the evidence of positive results, everyone felt different emotions and identified different positive aspects in this journey that had only just begun.

The thoughts of Agile team members who were part of the Agile transformation project are collected below (Table 2).

Based on the interviews conducted with various team members involved in Progetti e Soluzioni's Agile transformation, the advantages of using Agile methodology can be summarized as follows (Figure 5):

1. **Customer Centricity**: Agile methodology prioritizes delivering continuous value to customers by focusing on their needs and feedback. This customer-centric approach ensures that the product meets user requirements effectively.

A Product Owner said, "*After adopting this methodology at work, I have immediately had positive results: regular feedbacks at the end of each Sprint were valuable, because they allowed the stakeholders to give greater importance to what was a priority for them and the Scrum Team to keep working without having to analyze in detail the activities to perform.*

"*because customer centricity is fundamental. I would say that the three most impactful factors were:*

- *The development's primary aim to deliver continuous value to the customer;*
- *The use of an empirical approach to iterative development for implementing what was actually requested and necessary, in order to provide immediate feedback to the customer;*
- *The customer's regular involvement and feedback sharing in order to understand if what has been produced actually meets the needs of software users"*

2. **Iterative Development**: Agile methodology emphasizes iterative development, allowing for frequent feedback and adjustments throughout the development process. This iterative approach enables teams to respond quickly to changing requirements and market conditions.

A Developer said, "*After 20 years of software development in waterfall mode, I initially had many doubts, two in particular: How to ensure an effective software development without first making a full analysis of all the requirements, in the light of the waterfall approach that I had applied so far; How to best identify the MVP to implement for the target customers I had selected*".

"*But as far as I am concerned, I had an actual epiphany when I discovered what was the role of Story Points and how Stories and Sprints–as well as the project's following phases–were estimated. Contrary to what I had seen before, I understood that it is necessary to give up exact estimation in order to proceed with a heuristic approach in which Story Points are used as a relative measurement unit to assess the average return earned in the past and therefore to estimate future work*".

3. **Empowered Teams:** Agile methodology empowers cross-functional teams to make decisions and take ownership of their work. Teams collaborate closely with stakeholders, allowing for greater transparency, communication, and alignment.

A Product Owner said, "*The development method has inevitably changed since it had to adjust to the new framework. First of all, graphical interfaces have become a priority and, in some cases, they have allowed us to remove or edit some features before completing them on the backend side. This has saved us several days of work*".

4. **Flexibility and Adaptability**: Agile methodology promotes flexibility and adaptability, enabling teams to respond quickly to changing priorities and market demands. This flexibility allows for faster delivery of high-quality products.

A Developer said, "*After adopting this methodology at work, I have immediately had positive results: regular feedbacks at the end of each Sprint were valuable, because they allowed the stakeholders to give greater importance to what was a priority for them and the Scrum Team to keep working without having to analyze in detail the activities to perform*".

Another Developer said, "*The development method has inevitably changed, since it had to adjust to the new framework. First of all, graphical interfaces have become a priority and, in some cases, they have allowed us to remove or edit some features before completing them on the backend side. This has saved us several days of work*".

5. **Continuous Improvement**: Agile methodology fosters a culture of continuous improvement, where teams reflect on their work and identify opportunities for enhancement. This focus on continuous learning and adaptation drives innovation and growth within the organization.

An Agile Coach said, "*Throughout the pilot project's development, the company management was present at key moments and has actively participated in the training sessions, developing a*

shared language together with the Scrum Team. Learning and employing a shared language has led to an effective and efficient alignment between business and developers, avoiding the typical and dysfunctional "us-vs-them" mentality and making them always feel as they belonged to one same team".

A Developer said, "*Little by little, I have figured out that Scrum can be an extremely valid tool to manage complex or dematerialized projects and therefore it can be profitably implemented for software development*".

6. **Reduced Risk:** Agile methodology facilitates proactive risk management by breaking down projects into smaller, manageable increments. This approach allows teams to identify and address potential risks early in the development process, reducing the likelihood of project delays or failures.

A Developer said, "*It was not easy to estimate the amount of work that a Developer should be able to carry out, but–thanks to Product Backlog statistics–we are managing to become more aware of our velocity over time, without overestimating ourselves or underestimating difficulties and uncertainties. Sprint after Sprint, our estimations are becoming more and more accurate and negotiating with the PO less and less necessary".*

7. **Improved Collaboration and Communication:** Agile methodology encourages close collaboration and communication among team members, stakeholders, and customers. Regular meetings and interactions promote transparency, alignment, and shared understanding.

A Developer said, *"Usually, meetings are considered a waste of time, dragging on indefinitely and without any return. Otherwise, Scrum events are short and frequent, so they push the people involved towards continuous and productive confrontation. The brevity of the Sprint forces us to make frequent and close releases and therefore to get immediate and easily accepted feedback. The software is finally seen as a living product, which is continuously adapted to the needs of the stakeholders or the business owner and will never reach a final version after all".*

"It was rewarding to see how my relational and negotiation skills–gained over the years through managing customers and their requests–were extremely useful for a Product Owner, in addition of course to a thorough understanding of our customers and their operating field".

The Agile Coaches said, " *All the people involved have showed eagerness to work together and do it at their best, so they have quickly absorbed the skills I had transmitted, to the extent that I have observed they were put into practice in a systematic and very productive way".*

"At the end of the pilot project, the team was ready to proceed on its own and this is a great moment of personal and professional gratification for anyone who chooses to be an Agile Coach".

8. **Enhanced Product Quality**: Agile methodology prioritizes delivering working software in short iterations, allowing for early detection and resolution of issues. This focus on incremental delivery and continuous improvement leads to higher-quality products and increased customer satisfaction.

A Product Owner said, "*From a technical point of view, my software writing style has undergone some adjustments since I have become more aware of the fact that having a long-term scope is unacceptable. Now my code is more Object Oriented, in order to be scalable and maintainable. My planning, though, looks to the future but thinking short-term, so as to avoid having to erase entire modules–as it has often happened in the past–because they had become useless or totally inconsistent with the business owner new requirements".*

A General Manager said, *"Overall, the adoption of Agile methodology at Progetti e Soluzioni has resulted in a more responsive, collaborative, and customer-focused approach to software development, leading to improved outcomes and greater success in meeting user needs and business objectives".*

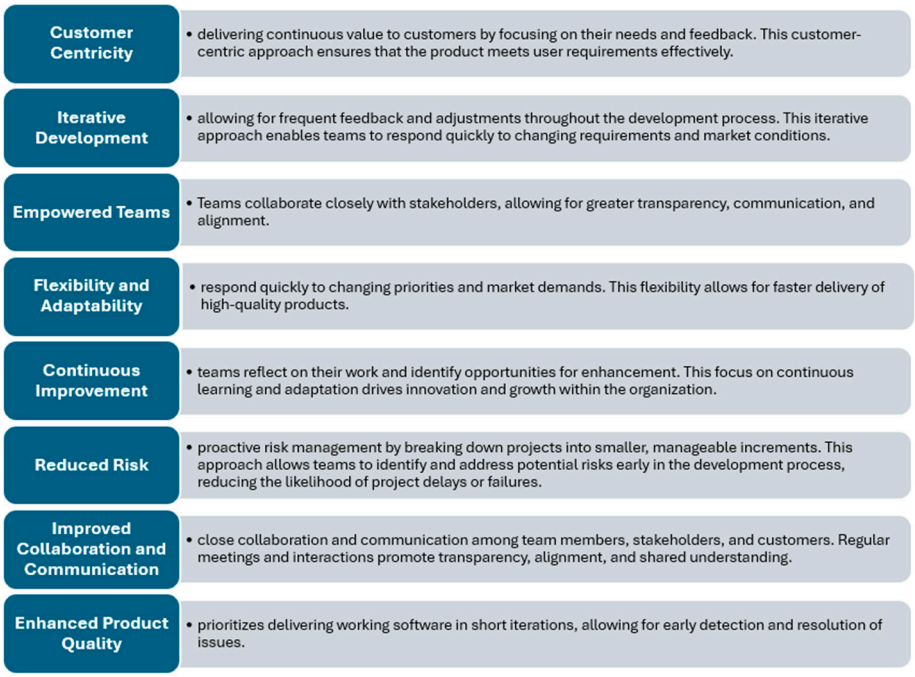


Figure 5. Interview results (own elaboration).

5. Conclusions and Discussion

This paper aims to respond to the following question: How can Agile transformations be successfully implemented in organizations to enhance Business Agility and address behavioral, cultural, and team-related challenges?

To respond to this question, we employed a longitudinal case study research (Yin 2018), based on three main phases: Framework proposal based on an Analysis of Problems Encountered; implementation and experimentation of the framework proposed during the duration of the project; interviews and evaluation post project.

The approach proposed and experimented to the case of Progetti and Soluzioni was proved to be effective and performant for implementing an Agile transformation process.

The ARTCO framework here proposed and tested represents the main practices that need to be employed to maintain the focus on people and to work always keeping in mind that the goal is to maximize value for the customer. By training the team for this attitude, it was possible to improve cooperative work and to establish an atmosphere of trust and transparency inside the team, whose members have faced uncertainties and change as an opportunity to test new solutions to make their customers happy. The team succeeded at dealing with conflicts through a win–win approach, being able to manage emotions and never losing sight of customer needs.

The people who have been part of this pilot project were undoubtedly essential to reach the goal. The adoption of the ARTCO framework helped the team rapidly become self-sufficient, and so it soon made the external Agile Coaches non-essential.

Takeaways that emerge from this research are as follows:

- Top Management Involvement: The active participation and support from top management were crucial. Their involvement from the inception of the project emphasized the strategic importance of the Agile transformation and fostered a high level of commitment from all stakeholders. The clear communication of the project's strategic value helped align organizational goals with the transformation effort, ensuring sustained support and resources.
- Institutionalization of Agile Transformation: By institutionalizing the pilot project, the company ensured sustained commitment and participation. The formal launch event and ongoing engagement created a shared sense of purpose and motivation

among team members. This structured approach provided a clear roadmap, making the transition smoother and more cohesive.

- Adaptation to Remote Work: The necessity for remote collaboration, enforced by the pandemic, presented challenges that were effectively managed through the Agile framework. The use of intuitive and familiar remote collaboration tools (Zoom, Whereby, Mural, Slack, Trello, and Google Sheets) facilitated seamless communication and collaboration, mimicking a physical office environment. The team's adaptability to these tools was critical in maintaining productivity and engagement despite physical separation.
- Frequent Interactions and Communication: The Agile framework's emphasis on frequent interactions reduced misunderstandings, enabled immediate responses to changes, and encouraged a high level of cooperation. This constant communication was vital in maintaining project momentum and alignment among team members. Regular meetings and feedback loops ensured that issues were promptly addressed, and everyone remained on the same page.
- Empowered and Motivated Teams: The Agile methodology empowered teams by promoting cross-functional collaboration and decision-making. Team members were encouraged to take ownership of their work, which increased their engagement and productivity. The sense of autonomy and responsibility fostered a proactive approach to problem-solving and innovation.
- Iterative Development and Customer Centricity: The iterative nature of Agile development allowed for continuous feedback and adjustments, ensuring that the product met customer needs effectively. This approach fostered a strong focus on delivering value to the customer. Regular involvement of customers and stakeholders in the development process ensured that the product evolved in line with user expectations and market demands.
- Continuous Improvement and Learning: The Agile framework promoted a culture of continuous improvement. Regular retrospectives allowed the team to reflect on their processes and identify areas for enhancement, driving innovation and growth. This iterative approach not only improved the product but also enhanced the team's skills and collaboration over time.
- Enhanced Product Quality: The focus on incremental delivery and frequent testing resulted in early detection and resolution of issues, leading to higher-quality products and increased customer satisfaction. The emphasis on quality and user feedback throughout the development process ensured that the final product was robust and aligned with user needs.

The proposed ARTCO framework, designed to maintain a focus on people and maximize customer value, proves to be an effective model for Agile transformation. These evidences are coherent with the study by Hoda and Noble (2017) that argues that Agile "take place within a multi-dimensional network of on-going changes in different areas of practice" (p. 10). Also, the study by Coimbra et al. (2023) highlights the crucial role of factors like people, culture, and leadership in successful Agile transformation. From a practical point of view, our study serves as a guide for project managers embarking on an Agile transformation journey. It also underscores the critical role of active top management participation and support in achieving successful Agile transformation. Formalizing Agile initiatives through structured approaches and continuous engagement fosters a shared sense of purpose among team members, ensuring sustained commitment and facilitating smoother transitions.

Limitations: As the study is based on a single case study, its results may not be highly generalizable. Also considering that the case study's focus on the IT sector, its broader applicability is limited. Therefore, future research could focus on Agile transformation cases that focus on other sectors. Further research across diverse settings is needed to validate the generalizability of the ARTCO framework. Secondly, the data collection was limited to eight interviews so a larger and more diverse sample could offer a more com-

prehensive understanding of the Agile transformation experience. ***Future studies*** could address these limitations by exploring diverse organizational contexts, using larger and more varied samples, and conducting long-term evaluations to gain a deeper understanding of the complexities involved in Agile transformations. Future research will focus on systematically gathering and analyzing data from the company to assess the impact of the Agile transformation. This ongoing activity aims to evaluate the effectiveness of implemented strategies, identify areas for improvement, and measure the overall outcomes on team performance and customer satisfaction. By establishing clear metrics and conducting follow-up interviews, we aim to provide a comprehensive understanding of the transformation's long-term effects and refine our approach based on empirical evidence. Also, future research could focus on a wider implementation and testing of the proposed framework in other contexts.

Author Contributions: Investigation, A.D.G.; Writing—review & editing, V.N. and A.I. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Informed consent was obtained from all subjects involved in the study.

Data Availability Statement: Data are contained within the article.

Conflicts of Interest: The authors declare no conflict of interest.

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